SECTION: 01

Question: 01

Analyze the algorithm to find best, Average, & worst Running Time:

Given algo is an optimized version of Bubble Sort.

→ Frequently executed operation:

```
A[j] > A[j+1];
```

(i) Best Case: When array is already sorted.

So $T_{(best)}(n) = n-1$

$$\Rightarrow T_{best}(n) = O(n)$$

(ii) Worst Case:

When array is reverse sorted.

Outer loop: $n-1$

Inner loop: $n-1$

Total: $(n-1)(n-1) \Rightarrow n^2-2n+1$

So,

$$T_{worst}(n) = O(n^2).$$

(iii) Average Case:

Expected.

$(n-1)^2$

which is Quadratic

So

$$T_{avg}(n) = O(n^2).$$

Question: 02

Compare Algo A & Algo B:

Algorithm A : $10n^2$

Algorithm B : $300 \log n$

Find n where B becomes better:

Sol:-

We need:

$$300 \log n < 10n^2$$

Divide by 10 $\Rightarrow$ $30 \log n < n^2$

Trying Values:

(i) $n = 4$:

$$30 \log(4) = 60$$

$$(4)^2 = 16$$

Not Satisfied.

(ii) $n = 8$:

$$30 \log(8) = 90$$

$$(8)^2 = 64$$

Not Satisfied.

(iii) $n = 10$:

$$30 \log(10) = 99.66$$

$$(10)^2 = 100$$

Satisfied.

Therefore $n \geq 10$

Algo B becomes faster from approximately $n = 10$ onwards.

Question: 03

Using Definitions of O & Ω

Show:

$$6n^2 + 20n \in O(n^3)$$

but

$$6n^2 + 20n \notin \Omega(n^3)$$

Sol:-

- Part A:

Need constants c & n_0 such that:

$$6n^2 + 20n \leq cn^3$$

for $n \geq 1$:

$$20n \leq 20n^3$$

Thus

$$6n^2 + 20n \leq 26n^3$$

Choose:

$$c = 26 \quad , \quad n_0 = 1$$

Therefore:

$$6n^2 + 20n \in O(n^3).$$

- Part B:

Assuming

$$6n^2 + 20n \in \Omega(n^3)$$

Then

$$6n^2 + 20n \geq cn^3$$

Divide by n^2:

$$6 + \frac{20}{n} \geq cn$$

As: $n \to \infty$

L.H.S tend to 6, while R.H.S to infinity.

Therefore:

$$6n^2 + 20n \notin \Omega(n^3)$$

Question: 04

Running Time Using Limits:

(i)

$f(n) = 3n^3 + 10n^2 + 5n + 100$ ⇒ Using L'Hopital Rule:

Sol: $g(n) = n^3$

Limit:

$$\lim_{n \to \infty} \frac{3n^3 + 10n^2 + 5n + 100}{n^3}$$

$$= 3$$

Finite Pos const-t

$$\boxed{O(n^3)}$$

(ii)

$f(n) = n\log n + n^2$

Sol: $g(n) = n^2$

⇒ $$\lim_{n \to \infty} \frac{n\log n + n^2}{n^2}$$

$$\lim_{n} \left(\frac{\log n}{n} + 1 \right)$$

$\Rightarrow 1$

Hence:

$$\boxed{O(n^2)}$$

(iii)

$$f(n) = 2^n + n^2$$

Sol:

$g(n) = 2^n$

$$\lim_{n \to \infty} \frac{2^n + n^2}{2^n}$$

$$\Rightarrow 1 + \frac{n^2}{2^n} \Rightarrow 1.$$

Therefore:

$$\boxed{O(2^n)}$$

(iv)

$$f(n) = \log^2 n + n \log n$$

Sol:

$g(n) = n \log n.$

$$\Rightarrow \lim \frac{\log^2 n + n \log n}{n \log n}$$

$$\Rightarrow \frac{\log n}{n} + 1$$

$\Rightarrow 1$

Hence:

$$\boxed{O(n \log n)}$$

Question: 05

Show Correctness of Given Statements:

a) $\lg n \in O(n)$

Need:

$$\lg n \leq cn$$

For $n \geq 1$:

$$\lg n \leq n$$

Choosing $c = 1$.

Therefore: $\lg n \in O(n)$. $\Rightarrow$ True.

b) $n \in O(n \lg n)$

Need:

$$n \leq c(n \lg n)$$

Div by n:

$$1 \leq c \lg n$$

For $n \geq 2$:

$$\lg n \geq 1$$

where $c = 1$.

$$n \in O(n \lg n). \Rightarrow \text{True.}$$

c) $n \lg n \in O(n^2)$

Need:

$$n \lg n \leq cn^2$$

Div by n:

$$\lg n \leq cn$$

True for large n. $\Rightarrow$ True

d) $\lg^3 n \in o(n^{0.5})$

Computing: $\lim_{n \to \infty} \frac{\lg^3 n}{n^{0.5}}$ $\Rightarrow 0$

Algo grows much slower than any positive power of n.

$\lg^3 n \in o(n^{1/2})$.

$\Rightarrow$ True.

Question: 06

Time Complexity of Nested Loops:

Given:

```
for (i=1; i<=n; i++) {
    j = n;
    while (j >= 1) {
        O(1);
        j = j/2;
    }
}
```

- Inner While loop.

Value of j: $n, n/2, n/4, n/8 \dots 1$.

after k iterations: $\frac{n}{2^k} \cdot 1$.

$2^k \cdot n \Rightarrow k \cdot \log_2 n$

Therefore: $O(\log n)$.

- Outer Loop:

  Runs: n times.

So, Total:

$$n \times \log n$$

Therefore:

$$T(n) = O(n \log n).$$

SECTION: 02

Question: 01

Solve the following Recurences:

a)

Iteration Method:

$T(n) = T(n-1) + cn, \quad n > 1$

$T(0) = 0$

Sol:-

1st $T(n) = T(n-2) + c(n-1) + cn$

2nd $T(n) = T(n-3) + c(n-2) + c(n-1) + cn$

$\vdots \quad \vdots \quad \vdots \quad \vdots$

kth $T(n) = T(n-k) + c[(n-k+1) + (n-k+2) + \cdots + n]$

Now, $T(0) = 0$

$n - k = 0 \Rightarrow k = n.$

Thus:

$T(n) = T(0) + c(1+2+3+\cdots+n)$

$T(0) = 0 \Rightarrow T(n) = c(1+2+3 \cdots + n).$

So: $1+2+3+\cdots+n = n(n+1)/2$

$T(n) = c\,\frac{n(n+1)}{2}$

$T(n) = \frac{c}{2}(n^2+n)$

Therefore:

$$T(n) = \Theta(n^2).$$

by

Substitution Method:

$$T(n) = T(n-1) + cn$$

From Iteration we founed.

$$T(n) = \frac{c}{2}(n^2+n).$$

Now Verifying:

Assume:

$$T(n-1) = \frac{c}{2}\left[(n-1)^2 + (n-1)\right].$$

$$T(n-1) = \frac{c}{2}\left[(n-1)^2 + (n-1)\right] + cn.$$

expanding squares:

$$(n-1)^2 = n^2 - 2n + 1$$

$$T(n) = \frac{c}{2}\left[n^2 - 2n + 1 + n - 1\right] + cn$$

$$= \frac{c}{2}\left[n^2 - n\right] + cn$$

Hence: $T(n) = \frac{c}{2}(n^2+n)$

& $T(n) = \Theta(n^2)$.

Question: 02

$$T(n) = 2T(n/5^{(i)}) + 6n^3$$

$T(1) = 6.$

Where n is a power of 5.

Sol:- Expanding:

$$T(n) = 2[T(n/25) + 6(n/5)^3] + 6n^3$$

$$= 2T(n/25) + \frac{12n^3}{125} + 6n^3$$

2nd $\quad T(n/25) = 2T(n/125) + 6(n/25)^3$

$$T(n) = 4T(n/125) + \frac{24n^3}{15625} + \frac{12n^3}{125} + 6n^3$$

kth $\quad T(n) = 2^k T(n/5^k) + 6n^3 \sum_{i=0}^{k-1} (2/125)^i$

Base Case: $\frac{n}{5^k} = 1 \Rightarrow n = 5^k$

$k = \log_5 n$

So,

$$T(n) = 2^{\log_5 n} T(1) + 6n^3 \sum_{i=0}^{\log_5 n - 1} \left(\frac{2}{125}\right)^i$$

Geometric Series:

$$\sum_{i=0}^{n-1} r^i = \frac{1-r^n}{1-r}$$

Since;

$$r = 2/125 < 1$$

Hence

$$T(n) = \Theta(n^3).$$

(ii)

$$T(n) = 40T(n/3) + 2n^3$$
$$T(1) = 2$$
n is a power of 3.

Sol:-

$$T(n/3) = 40T\left(\frac{n}{9}\right) + 2\left(\frac{n}{3}\right)^3$$

Subtitute back

1st $$T(n) = 40\left[40T\left(\frac{n}{9}\right) + 2\left(\frac{n}{3}\right)^3\right] + 2n^3$$

$$T(n) = 40^2T\left(\frac{n}{9}\right) + 80\left(\frac{n^3}{27}\right) + 2n^3$$

$$T(n) = 40^2T(n/9) + \frac{80}{27}n^3 + 2n^3$$

2nd $$T(n/9) = 40T\left(\frac{n}{27}\right) + 2\left(\frac{n}{9}\right)^3$$

$$T(n) = 40^2T\left[40T\left(\frac{n}{27}\right) + 2\left(\frac{n}{9}\right)^3\right] + \frac{80}{27}n^3 + 2n^3$$

$$T(n) = 40^3T\left(\frac{n}{27}\right) + \frac{3200}{729}n^3 + \frac{80}{27}n^3 + 2n^3$$

3rd $$T(n/27) = 40T\left(\frac{n}{81}\right) + 2\left(\frac{n}{27}\right)^3.$$

Subtitute.

$$T(n) = 40^4T\left(\frac{n}{81}\right) + \frac{128000}{19683}n^3 + \frac{3200}{729}n^3 + \frac{80}{27}n^3 + 2n^3$$

Pattern: $2n^3, 2n^3(40/27), 2n^3(40/27)^2, 2n^3(40/27)^3$.

k^{th} $T(n) = 40^k T\left(\frac{n}{3^k}\right) + 2n^3 \sum_{i=0}^{k-1} \left(\frac{40}{27}\right)^i$

When $n/3^k = 1$

$3^k = n$, $\boxed{k = \log_3 n}$

So,

$$T(n) = 40^{\log_3 n} T(1) + 2n^3 \sum_{i=0}^{\log_3 n - 1} \left(40/27\right)^i$$

$T(1) = 2$.

Geometric Series:

$r = 40/27 > 1$

Largest term dominates.

Last term is $\Rightarrow n^3 (40/27)^{\log_3 n}$

identity:

$$a^{\log_b n} = n^{\log_b a}$$

$$= n^3 n^{\log_3 (40/27)}$$

$$= n^{\log_3 40}$$

Hence:

$$T(n) = \Theta\left(n^{\log_3 40}\right)$$

$$T(n) = \Theta\left(n^{3.357}\right)$$

Question: 03

Solve the Recurrences Using Theorem:

(i)

$$T(n) = 15T(n/5) + 6n$$

$$T(1) = 1$$

n is the power of 5.

Sol:-

$$T(n) = aT(n/b) + f(n)$$

We get

$$a = 15$$

$$b = 6$$

$$f(n) = 6n$$

- Computing exponent:

$$n^{\log_b a}$$

$$n^{\log_5 15}$$

So,

$$\log_5 15 = \frac{\log 15}{\log 5} \approx 1.683$$

$$n^{\log_5 15} = n^{1.683}$$

- Comparing Growth rates:

$$f(n) = 6n = \Theta(n)$$

and

$$n^{\log_5 15} = n^{1.683}$$

$$n < n^{1.683}$$

So, recursion dominates.

$$T(n) = \Theta(n^{\log_2 1.5})$$

$$\boxed{T(n) = \Theta(n^{1.585})}$$

(ii)

$$T(n) = 2T(n/3) + cn^5$$

Sol:-

$$T(n) = aT(n/b) + f(n).$$

$a = 2$

$b = 3$

$f(n) = cn^5$

- Computing: $\log_3 2 \approx 0.631$

we get. $n^{\log_3 2} = n^{0.631}$

- Comparing: $f(n) = cn^5$

and $n^{0.631}$

Clear: $n^5 > n^{0.631}$

Non-recursive work dominates.

• Check:

verifying $af(n/b) \leq kf(n)$.

for some $k < 1$

$$2c(n/3)^5 = \frac{2c}{243} n^5$$

Since $2/243 < 1$.

$$T(n) = \Theta(n^5).$$

Question: 05

Using Master Theorem:

$$T(n) = 5T(n/3) + cn^8 + cn \quad \text{(i)}$$

Sol:

$$= aT(n/b)$$

$a = 5$

$b = 3$.

- Computing:

$$n^{\log_3 5}$$

$\log_3 5 \approx 1.465$.

Thus $n^{\log_3 5} = n^{1.465}$.

- Dominant Term in $f(n)$.

$cn^8 + cn \Rightarrow cn^8$

- Comparing:

n^8

with

$n^{1.465}$

Clearly, $n^8 \gg n^{1.465}$

So, $T(n) = \Theta(n^8)$.

(ii)

$$T(n) = 5T(n/4) + cn^3 + cn^2$$

Sol:-

$a = 5$

$b = 4$

- Computing:

$$n^{\log_4 5}$$

$$\log_4 5 \approx 1.161$$

- cn^3

Compare n^3 with $n^{1.161}$

clearly: $n^3 \gg n^{1.161}$

Case 3.

$$T(n) = \Theta(n^3).$$

(iii)

$$T(n) = 4T(n/4) + 2n^2 \log n$$

Sol:-

$a = 4$

$b = 4$

- $n^{\log_4 4} = n$

- Comparing:

$f(n) = 2n^2 \log n$

with

n

clearly

$2n^2 \log n \gg n$

Case 3 applies.

- Regularity Condition:

$$4f(n/4)$$
$$= 4[2(n/4)^2 \log(n/4)]$$
$$= 1/2\, n^2 \log(n/4)$$

It's less than:

$$k(2n^2 \log n)$$

for some $k < 1$

condition holds.

$$T(n) = \Theta(n^2 \log n).$$